

ASSIGNMENT # 01

Subject: Artificial Intelligence
Program: MCA
Due Date: 2-10-2024

Sub Code: 22MC314
Semester: 03 (Batch-2023)
Full Marks: 50

Answer all 10 Questions. Each question carries 5 marks.

Q No.	CO*	Level	(1) Remembering (2) Understanding (3) Applying (4) Analyzing (5) Evaluating (6) Creating																		
1	1	1	Describe the attributes of an Intelligent agent. Explain different types of environments in which agents have to operate.																		
2	1	2	Distinguish between Reactive and Deliberative agents in terms of their structure and functionality. Give examples of each type of agents.																		
3	1	5	How is a Goal-driven agent different from Utility-driven agent? Justify their suitability for different application areas.																		
4	2	6	Model 8-puzzle as a state space search problem. Create a State space to reach the Goal state GS from a given Initial state IS as shown below. Expand upto 4 levels of the search tree. <table><tr><td>1</td><td>8</td><td>7</td></tr><tr><td>2</td><td></td><td>6</td></tr><tr><td>3</td><td>4</td><td>5</td></tr></table> IS → <table><tr><td>1</td><td>2</td><td>3</td></tr><tr><td>4</td><td>5</td><td>6</td></tr><tr><td>7</td><td>8</td><td></td></tr></table> GS	1	8	7	2		6	3	4	5	1	2	3	4	5	6	7	8	
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4	5	6																			
7	8																				
5	2	4	Perform a critical analysis of Iterative Deepening and Iterative Broadening Search techniques with their applicability.																		
6	3	3	What is the role of heuristics in informed search techniques? Explain the A* algorithm with an example.																		
7	3	6	State the 3 Missionaries and 3 Cannibals trying to cross a river as a Constraint Satisfaction problem. Provide a possible solution to the problem.																		
8	2	5	What is the significance of Adversarial Search? How does alpha-beta pruning help?																		
9	2	2	What is the role of predicate logic in representing knowledge for AI systems? How does it help in inferring new knowledge?																		
10	2	6	Explain the use of Semantic net in knowledge representation. Develop a property inheritance algorithm to retrieve the value of an attribute for an object belonging to a particular class.																		